Odoo x Gujarat Vidyapith Hackathon '25

Team Name:

KissanVarg

Team Members:

Rahul Birawat – Full stack Developer (+ Blockchain Integration)

Abhinav Choukse – Frontend Developer + UI/UX Designer

Meet Parmar – Backend Developer

Het Patel – Project Manager, Business Strategy

Problem Statement

Chosen Problem:

Bridging Natural Farmers and Conscious Consumers

Problem Analysis:

Natural farmers struggle to find reliable markets due to:

- **Limited Market Access**: Dependence on intermediaries leads to reduced income.
- **Lack of Trust & Recognition**: Consumers find it hard to differentiate natural produce due to a lack of proper certification.
- **High Certification Costs & Complexity**: Organic certification is expensive and time-consuming, making it inaccessible for small farmers.

Consumers face challenges such as:

- **Lack of Transparency**: Difficulty verifying if a product is genuinely natural due to mislabeling and greenwashing.
- **No Direct Connection to Farmers**: Consumers rely on third-party vendors without assurance of authenticity.

Target Audience:

- Natural & Organic Farmers
- Conscious Consumers

Solution Overview

Brief Explanation:

Our solution is a **digital marketplace** connecting farmers directly with consumers, ensuring trust, transparency, and fair trade.

Approach:

- **Direct Market Connection**: A platform where natural farmers list products and transact directly with consumers.
- **Farmer Authentication System**: We verify government-issued certification and display farmers' certification levels (Level 1 - In conversion organic, 2 - Certified Organic, or 3 - PGS Certified Natural Farmer) on their profiles.
- **Affordable Digital Certification**: AI-driven low-cost certification in partnership with government agencies.
- **QR Code-Based Consumer Traceability**: Consumers scan QR codes to verify authenticity, farmer details, and pricing.
- **Community Engagement & Trust-Building**: Features such as direct farmer-consumer messaging, pre-ordering, and storytelling.
- **Farming Practice Transparency:** Each farmer's irrigation, soil management, fertilization, and pest control methods are recorded and displayed to consumers.
- **Crop & Cultivation Details:** Consumers can view crop type, seed variety, source, and sowing-to-harvest timeline for complete traceability.
- **Harvesting & Post-Harvest Info:** Displays harvesting method, storage conditions, processing type, and eco-friendly packaging details.
- **Helping farmers by Donation:** A page that will help farmers get donations and support if natural calamities destroy their crops.

Uniqueness:

- Blockchain-powered certification ensures tamper-proof records.

- AI-driven certification system reducing costs and complexity.
- Direct farmer-consumer connection fostering trust and fair pricing.

Frameworks/Technologies

Frontend/UI:

- React JS with Vite (Fa velopment, optimized performance)
- Tailwind CSS (Minimal and responsive styling)
- Material UI & Daisy UI (Prebuilt standard components)

Backend:

- Node.js & Express.js (API creation and management)
- Postman (API testing)

Database:

- MongoDB (Storage for farmers, consumers, and product data)

Blockchain:

- Etherscan (QR Code-based verification)
- NFTs (Digital certification mechanism)

Deployment:

- Render & Vercel (Web app hosting)
- GitHub (Version control)
- Docker (Containerization)

Reasoning for Technology Choices:

- **Scalability**: MongoDB and blockchain ensure smooth scalability.
- **Security**: Blockchain prevents data tampering.
- **Performance**: React + Vite ensures fast rendering.
- **Cost-effectiveness**: Open-source tools reduce development costs.

Assumptions & Constraints:

- Farmers must have access to smartphones.

- Government and NGO support for certification validation.
- Internet connectivity for seamless transactions.

Feasibility and Implementation

Implementation Ease:

- The marketplace can be developed as a **web-based application**, requiring minimal resources.
- Blockchain integration ensures long-term **security** and **trust**.

Effectiveness:

- Ensures **direct trade** between farmers and consumers, reducing dependency on intermediaries.
- Provides **tamper-proof certification** through blockchain.
- Encourages **community engagement** with farmer stories and direct interactions.

Business Scope and Use Case

Use Case Scenarios:

- A farmer lists organic tomatoes, receives direct orders, and gets verified by blockchain.
- A consumer scans a product's QR code at a local store and verifies authenticity.
- An NGO partners with the platform to sponsor farmer certifications.

Market Need:

- Growing demand for **organic and natural produce**.
- Increasing consumer awareness of **food sourcing transparency**.
- Lack of existing platforms ensuring **direct farmer-to-consumer transactions**.

Revenue Model (Optional):

- **Subscription Model**: Consumers subscribe to monthly fresh produce deliveries.
- **Transaction Fee**: Small percentage per sale for platform maintenance.
- **Government & NGO Partnerships**: Support for certification initiatives.

System Design and Architecture

Technologies Overview:

- **React + Vite** for frontend
- **Node.js + Express.js** for backend
- **MongoDB** for database
- **Blockchain (NFT-based certification)**

Design Patterns:

- **MVC (Model-View-Controller)** for frontend-backend separation.
- **Microservices architecture** for scalability.

Functional Flow:

1. Farmer submits produce details → Product gets verified.
2. Consumer scans QR code → Authenticity details retrieved from blockchain.
3. Purchase completed → Transaction recorded on the database.

Coding Approach

Development Strategy:

- **Agile methodology** for iterative development.
- **Modular design** for easy feature expansion.

Coding Standards:

- Follows **RESTful API principles**.
- Uses **code reviews & automated testing**.
- Maintains **well-documented codebase** for collaboration.